Lifting dfranzen's ARC-AGI-3 harness: what moved the hidden-game score

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Abstract

We tested 26 changes on top of dfranzen's Milestone 2 notebook for the ARC Prize 2026 ARC-AGI-3 Kaggle competition (offline, 9 hours, one RTX PRO 6000, 110 hidden interactive games, RHAE score). Changes were screened on a 25-game local run, on a 40-episode stuck-level suite, and in 33 public submissions. The D′ slot priority from a public notebook and plain dfranzen are within noise: 27.37 (8 draws) against 25.56 (5 draws), with draws of identical code spread over 22–30. One change cleared every draw of the kernel without it: solved-level memory, which pins the model's own one-line rule for each cleared level into the system prompt, scored 32.64 and 28.01 against 25.53 over five draws of the same kernel without it (Welch $t = 1.97$, $n = 2$). Our best public score, 32.64, ranks 296 of 5,070.

1 Setup

The base is dfranzen's notebook: Qwen3.8-Flash-Next (W4A16) with a multi-token-prediction drafter, served by SGLang "Pennyroyal" 2.5.3, driving a Python-tool agent harness over 10 concurrent game streams with a priority scheduler. Kaggle allows one submission per day; the public score covers about 55 of the 110 hidden games.

Three screens, with their measured noise:

- **Public score**: identical plain copies scored 28.39, 27.86, 24.60 and 25.21; with D′'s four draws the sd is about 2.8; community copies average 25.77 (sd 3.93).
- **Local 25-game run** (RHAE on the public games): identical runs differ by up to about 7 points.
- **Stuck-level suite**: 20 levels where full runs got stuck, each replayed twice from the recorded history; identical baselines cleared 9 and 9 of 40 episodes (sd about 3.7), so a gain needs about 16.

2 Funnel

Stage	Count
Mechanisms built	26
Tested on the 25-game local run	17
Tested on the stuck-level suite	12
Above the local noise band (42.9 ± 7)	1
Reproduced when stacked or re-run	0
Submitted publicly (as changes on dfranzen)	8
Beat plain dfranzen's public mean	2
Significant against plain ($p < 0.05$)	0

3 Measurements

Public submissions

Date	Configuration	
10-02	dfranzen + level prior + fade	
10-03, 10-04, 10-05, 10-13, 10-15	plain dfranzen copy	28.39, 27.86, 24.60,
10-06 . . . 10-19 (8 draws)	D′ slot priority (shiiin9)	29.68, 29.71, 26.60, 22.40, 29.85, 24.45,
10-08	dfranzen + endgame FIFO rotation	
10-10	D′ + re-attempt, new kernel slug	0.00 (r
10-11	D′ + re-attempt, proven kernel	
10-17	plain, our own GPU run overlapping	0.00 (r
10-20	D′ + 256 px + retrained drafter (v5)	
10-21 . . . 10-24	D′ and v5, slow weight loads	0.00, 0.00, 0.0
10-25, 10-28 . . . 10-31	D′ + keep-alive (v6)	23.90, 25.26, 25.61,
10-26, 10-27	v6 + turn archive (v7)	
11-01, 11-02	v6 + solved-level memory (v8)	3

Local 25-game runs

Change	Score	Levels	tok/s
unmodified (mean of 3)	42.9	101	597
re-attempt v4 (keeps slot)	50.18	109	578
HiCache, 160K context	48.63	108	530
level prior + fade (2 runs)	48.6	104	–
endgame FIFO rotation	47.88	109	–
HiCache 160K + re-attempt	42.56	98	547
256 px board images	42.70	99	627
re-attempt v3 (lost its slot)	42.31	95	542
24K history drain	44.93	97	416
reasoning window	24.27	74	544

Stuck-level suite (episodes cleared of 40; baseline 9, 9)

Stall probe 10; prompt candidates 11, 11, 12, 9, 5; win records 8; thinking cap 8; keep-context restart 8; replay-verified simulator check 8 (the contract reached all 40 transcripts, none wrote the `predict` function it asked for).

Re-attempt conversion

Of 28 re-attempts (v3 and v4), 6 cleared their level afterwards (21%), against about 25% of stuck levels cleared by continuing in unmodified runs.

Serving speed

Speculative-decoding settings, 100 real requests at 10 concurrent: dfranzen's 3 draft steps with the FR-Spec vocabulary gave 400 and 419 tok/s on two boots; 2 steps 393, FR-Spec off 392, no speculative decoding 312. 4 and 5 steps did not boot within 20 minutes. Replaying the scheduler over three local runs with a throughput multiplier gave 50.5 (stock), 52.4 (+10%), 54.3 (+20%) and 57.2 (+30%); the replay overstates the stock run (50.5 against 42.7 measured), so these are direction and rough size.

Phase 2 changes (14 October–2 November)

Keep-alive. From 10-20 scored machines took 35–63 minutes to load weights, longer than the 15 minutes Kaggle allows a scorecard without an action; three draws scored 0.00 and one errored. A watchdog cell RESETs one live game whenever no action has gone out for 5 minutes while the server loads (10 once it is up). All nine draws after it scored.

Turn archive. The model's own trimmed turns, searchable from the Python tool. Two draws (26.41, 24.39) fell under the gate fixed beforehand (two-draw mean ≤ 27 drops it).

Solved-level memory. Taken from a public notebook (vickyfym, after rellik13/sirikilohit, CC-BY 4.0). When a level clears, the next two turns ask for one "Rule for level N:" line; the rule, the winning attempt's last 30 actions and the clearing turn's reasoning are pinned at the end of the system prompt, rewritten only when the conversation drains so the prompt cache survives. Its save run (one game, 45K tokens) captured four rules with no errors but never drained, so our gate (at least one pin) failed. We overrode the gate after seeing why; the two public draws are the first full-length test.

4 Findings

1. Scheduler changes that move time between games won locally and lost publicly (20.60, 22.56): the 25-game local mix does not represent the 110 hidden games.
2. The model depends on its own past reasoning: removing it collapsed the run to 24.27.
3. Restarting a stuck level, with or without its context, converts it no more often than continuing.
4. Smaller board images (256 px) cost nothing and add 5% throughput. dfranzen's speculative-decoding settings are already the fastest tested; the remaining speed lever is a drafter retrained on the quantised model.
5. Asked to write a checkable simulator of the game, the model declined in every episode and called it optional: it predicts in prose, not in code.
6. A submission from a newly created kernel scored 0.00 although its save run passed; the same code from an existing kernel scored normally.
7. The model's own stated rules carried across a drain lifted the score where every attempt to carry raw history, records or notes had not: 30.33 over two draws against 25.53.
8. Late in the competition infrastructure, not policy, decided draws: five of eight draws from 10-17 to 10-24 failed in the scored rerun before the keep-alive.

5 What is not established

- Whether solved-level memory is a real gain: two draws, and 28.01 sits inside the range of the kernel without it.
- The private score, which Kaggle withholds until the close; public covers about 55 of the 110 hidden games.
- Whether D$'$ beats plain dfranzen: 27.37 against 25.56; resolving 2 points needs about 60 draws per arm.
- Whether a 10–15% throughput gain shows publicly: the replay puts it near +1–1.5 points, below what eight draws per arm can detect.

- Any local effect under about 5 points.
- The cause of the new-kernel 0.00: no scored-rerun logs are available; a Python 3.13 image change, a dataset-version race and mount failures all fit.
- What the teams above 40 do: the top three have not published code.
- Whether 256 px images change the public score: the one draw with them (and the retrained drafter) scored 24.10.

Sources

Results: `docs/experiments/arc-agi3-ledger.json`. Dashboard: `https://claude.ai/artifact/LsbFyTwgMn4EmDU2vN8mko`. Literature: `reports/ARC AGI 3 SOTA October update.md` and `reports/ARC AGI 3 SOTA October 12 delta.md`. Write-up of the final weeks: `docs/experiments/2026-11-02-arc-agi3-final`